Customer Analytics Platform

Churn Prediction, Customer Segmentation & Lifetime Value Forecasting

Results Report

Dataset: IBM Telco Customer Churn | 7,043 customers

Executive Summary

This project builds an end-to-end customer analytics pipeline on the IBM Telco Customer Churn dataset, combining customer segmentation, churn prediction, and customer lifetime value (CLV) forecasting into a single workflow. Model outputs are translated into prioritized, rule-based business actions so retention spend can be targeted at the customers who matter most.

Headline results:

- Churn rate across the customer base is **26.5%** (1,869 of 7,043 customers).
- Best churn classifier: **LightGBM** — F1-Score 0.571, ROC AUC 0.840.
- Best CLV regressor: **Random Forest Regressor** — $R^2 \approx 1.000$, RMSE 10,556.5.
- K-Means clustering identifies two clear segments: a High-Value group and an Inactive group.
- A rule-based engine combines segment + churn risk + CLV into a Critical/High/Medium/Low campaign priority.

Dataset Overview

The IBM Telco Customer Churn dataset contains 7,043 customers and 21 raw attributes covering demographics, account information, subscribed services, and billing. The target variable for classification is **Churn** (Yes/No); an **Estimated_CLV** value is derived for the regression task.

Metric	Value
Customers	7,043
Raw features	21
Churned customers	1,869 (26.5%)
Retained customers	5,174 (73.5%)
Avg. tenure	32.4 months
Avg. monthly charges	$64.76

Exploratory Data Analysis

Before modeling, the data was examined for churn distribution, spending patterns, and contract behavior. Customers on month-to-month contracts churn at a far higher rate than those on one- or two-year contracts, and churned customers tend to pay higher monthly charges.

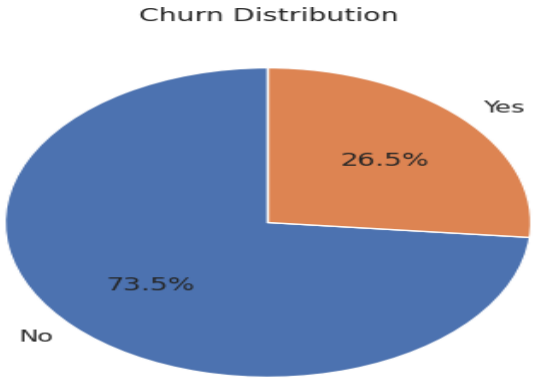


Figure 1. Overall churn distribution (26.5% churn rate).

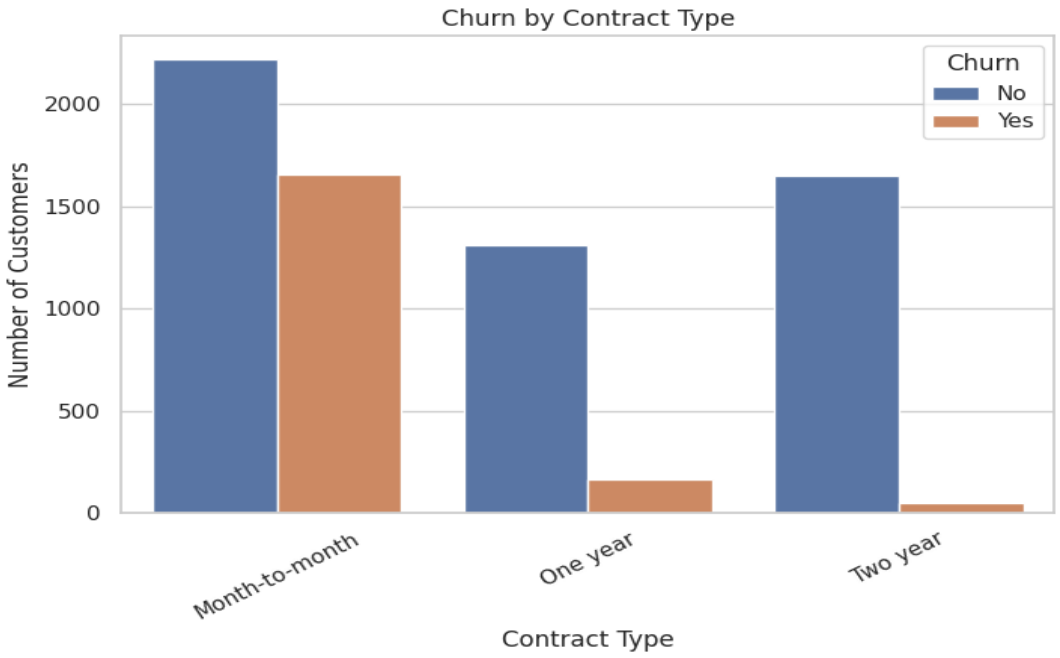


Figure 2. Churn counts broken down by contract type — month-to-month customers churn disproportionately.

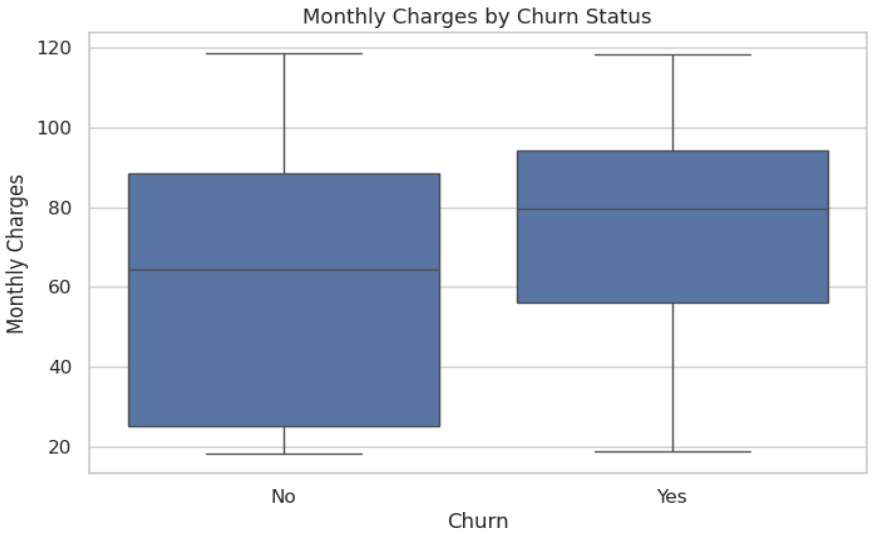


Figure 3. Monthly charges distribution by churn status — churned customers skew toward higher bills.

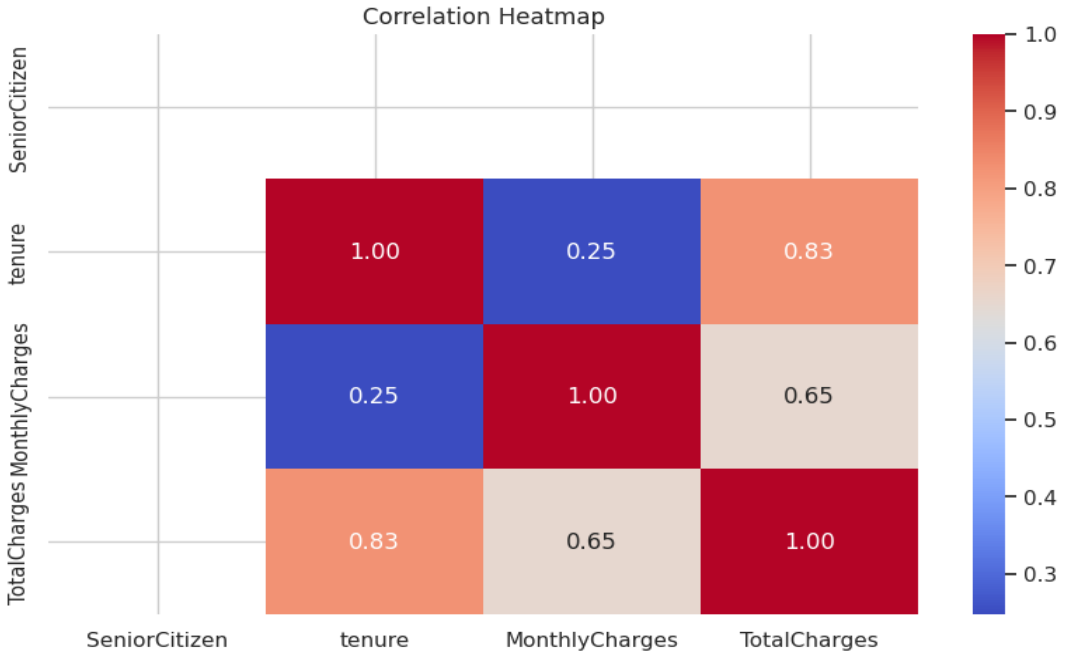


Figure 4. Correlation matrix of numeric features.

Feature Engineering

Raw fields were transformed into business-oriented features adapted from classic RFM (Recency / Frequency / Monetary) analysis for a subscription context:

Feature	Description
Customer_Tenure	Proxy for recency / loyalty
Monthly_Spending, Monetary_Value	Recurring and total value
Average_Order_Value	Average transaction size
Total_Transactions / Purchase_Frequency	Engagement volume
Product_Diversity	Number of services subscribed (“stickiness”)
Engagement_Frequency	Tenure vs. product-diversity ratio
Avg_Revenue_per_Month, Tenure_Buckets, Has_Tech_Support	Additional business-driven features

Customer Segmentation

RFM analysis grouped customers into actionable segments (Champions, Loyal Customers, Potential Loyalists, New Customers, At-Risk Customers, Lost Customers). K-Means clustering was then applied to the engineered feature set, with the optimal number of clusters (k=2) chosen via the Elbow Method and Silhouette Score. Hierarchical clustering was run as a cross-check.

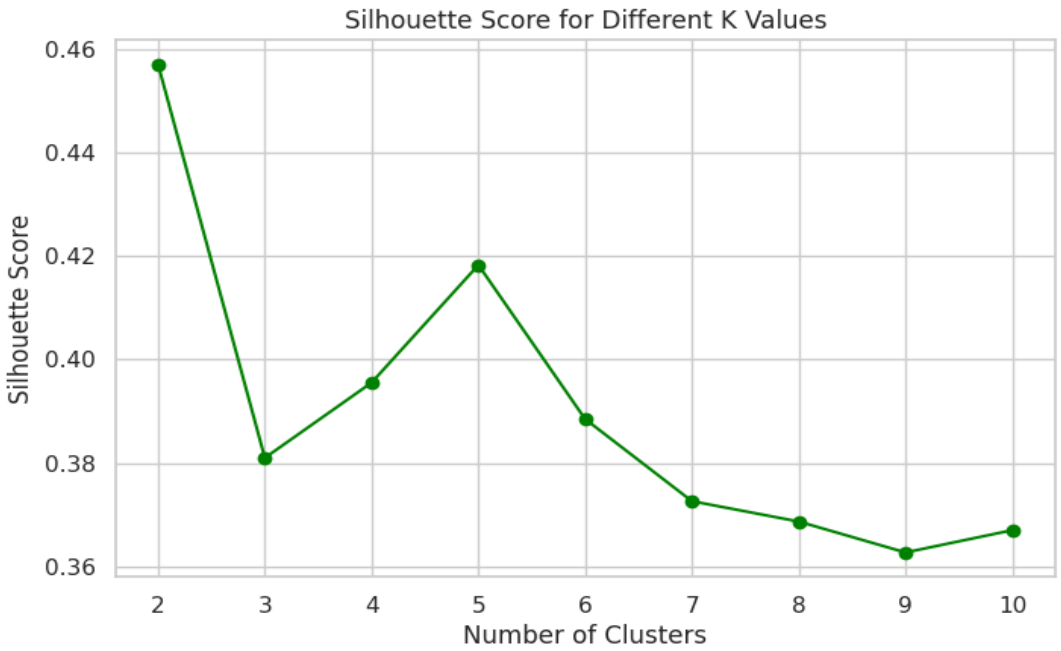


Figure 5. Silhouette score vs. number of clusters — best k = 2.

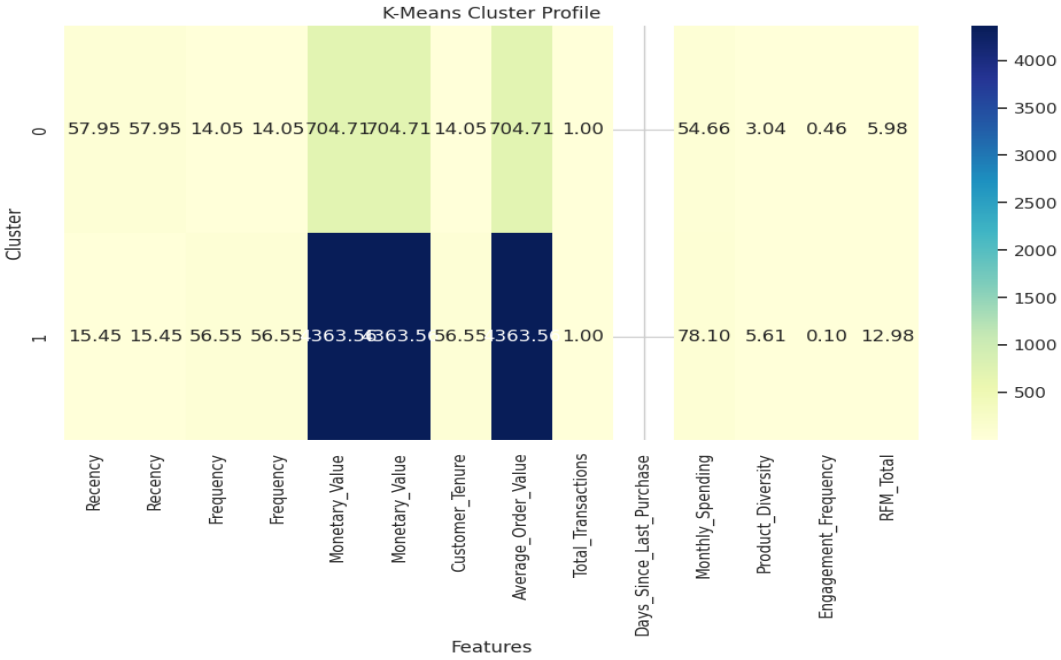


Figure 6. Average feature values per K-Means cluster: Cluster 0 (Inactive) vs. Cluster 1 (High Value).

Cluster 0 — Inactive Customers: higher recency, lower frequency and monetary value, lower tenure and engagement. **Business action:** re-engagement campaigns and offers to reduce dormancy.

Cluster 1 — High Value Customers: recent activity, high frequency and monetary value, longer tenure, higher product diversity. **Business action:** loyalty programs, upselling, and referral incentives.

Churn Prediction

Five classifiers were trained and compared on a held-out test set:

Model	Accuracy	Precision	Recall	F1 Score	ROC AUC
LightGBM (best)	**0.802**	**0.669**	**0.499**	**0.571**	**0.840**
XGBoost	0.804	0.686	0.480	0.565	0.843
Logistic Regression	0.791	0.654	0.450	0.533	0.842
Random Forest	0.767	0.574	0.458	0.510	0.802
Decision Tree	0.732	0.493	0.491	0.492	0.658

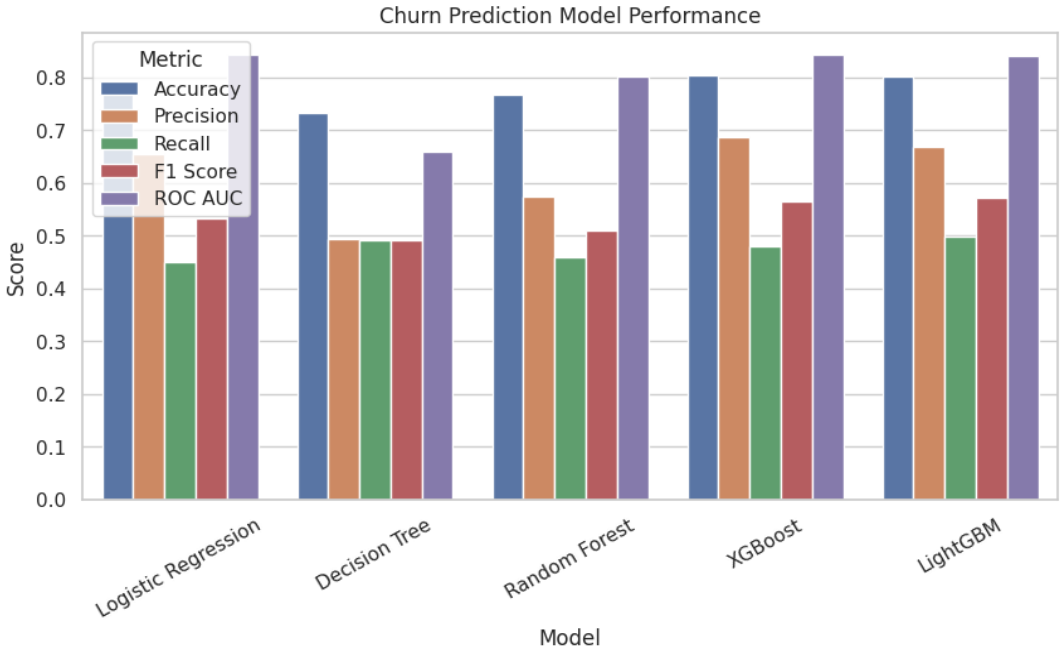


Figure 7. Churn model comparison across accuracy, precision, recall, F1, and ROC AUC.

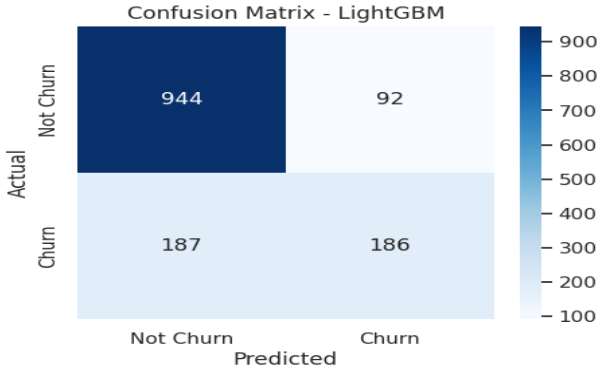


Figure 8. Confusion matrix for the best model (LightGBM).

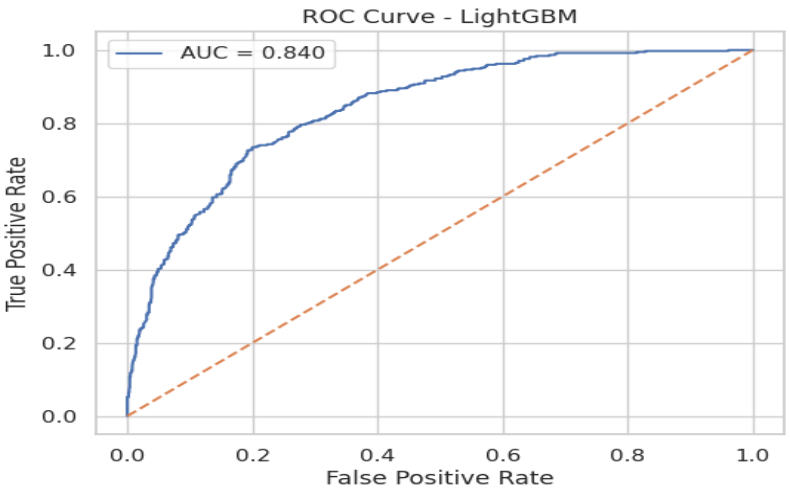


Figure 9. ROC curve for the best model (AUC = 0.840).

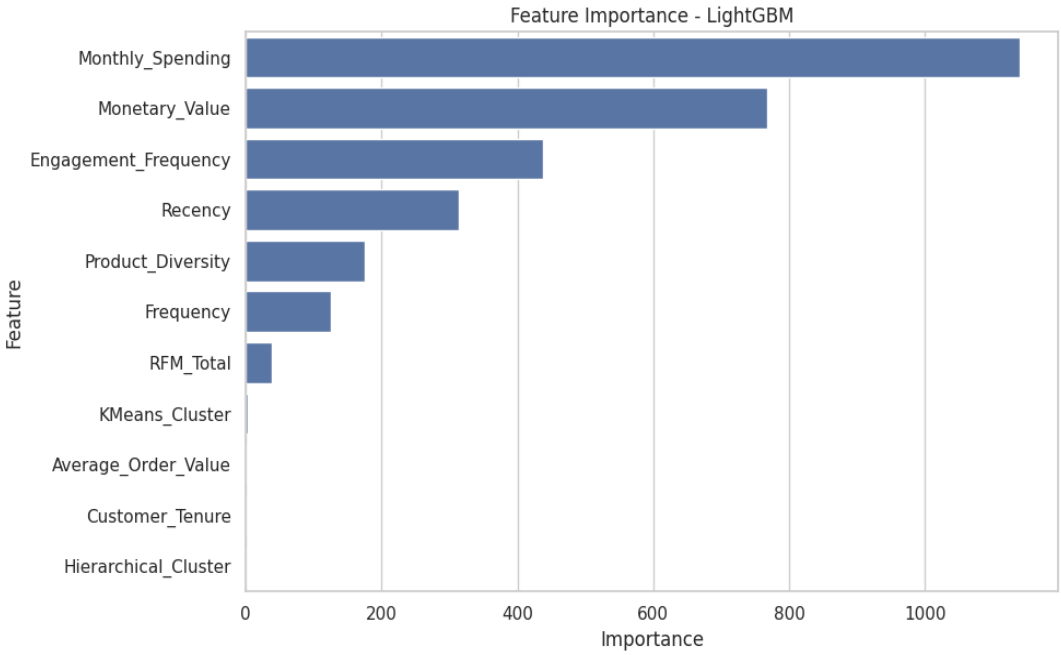


Figure 10. Feature importance (LightGBM) — Monthly_Spending and Monetary_Value dominate.

Takeaway: LightGBM was selected as the best model by F1-Score, balancing precision and recall on an imbalanced target. Monthly spending, monetary value, engagement frequency, recency, and product diversity are the strongest churn signals — aligning with the intuition that disengaged, lower-activity customers churn more.

Customer Lifetime Value (CLV) Prediction

Three regressors were trained to predict Estimated_CLV from engineered features:

Model	MAE	RMSE	R-Squared
Random Forest Regressor (best)	**4,886.9**	**10,556.5**	**1.000**
Gradient Boosting Regressor	22,316.3	40,622.3	0.998
Linear Regression	178,990.1	325,397.6	0.899



Figure 11. CLV regression model comparison by RMSE (lower is better).

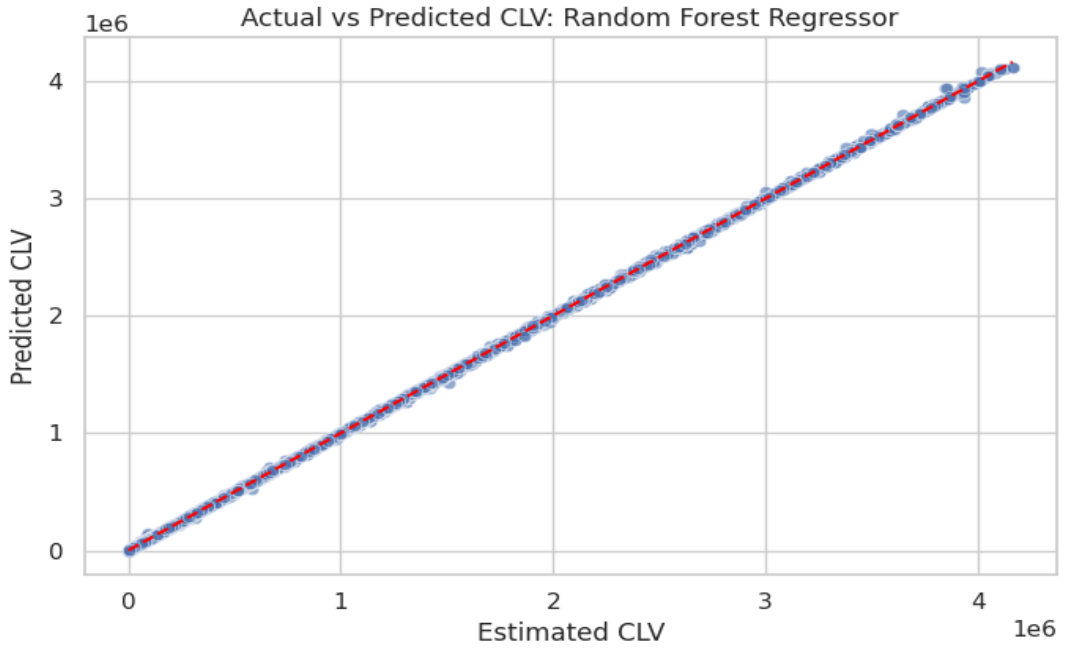


Figure 12. Actual vs. predicted CLV for the best model (Random Forest Regressor).

Takeaway: The Random Forest Regressor achieved near-perfect fit, allowing customers to be reliably segmented into Low / Medium / High CLV tiers for downstream targeting.

Business Recommendation Engine

RFM segment, churn probability, and predicted CLV are combined into a rule-based recommendation and a Campaign_Priority label, so retention budget is pointed at the customers who matter most.

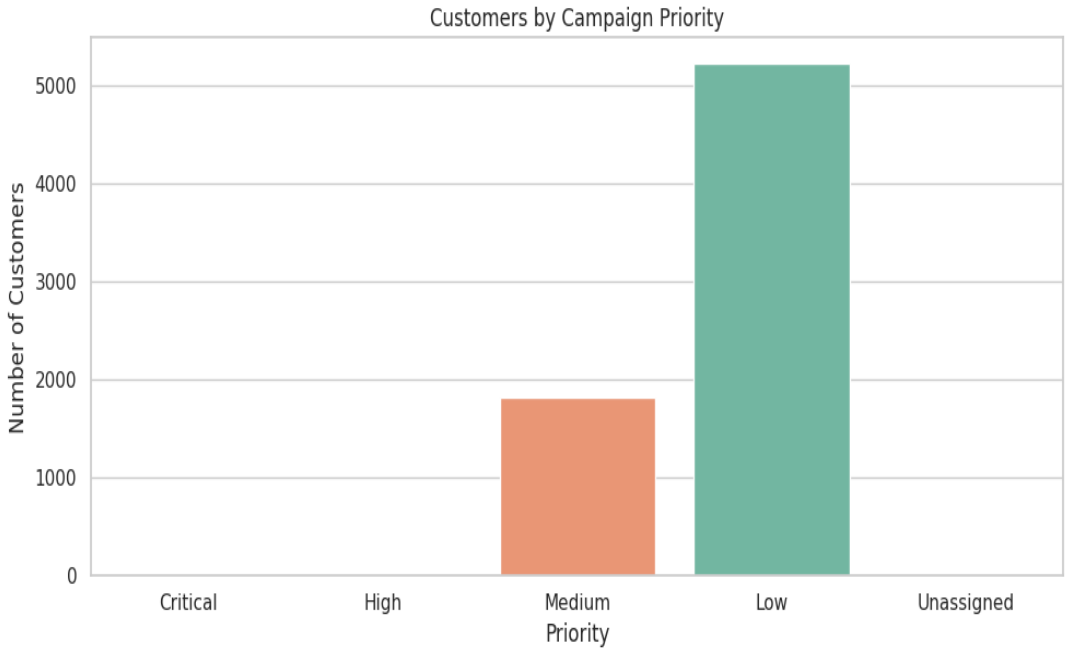


Figure 13. Customers grouped by campaign priority (Critical / High / Medium / Low).

Priority	Trigger	Recommended Action
Critical	High CLV + high churn risk	Personalized discount, priority retention outreach
High	Lost/dormant or at-risk segment	Reactivation campaign or retention offer + feedback
Medium	High purchase frequency, Champions/Loyal, or Loyal potential	Loyalty reward, upsell, or onboarding
Low	Low CLV + low churn risk	Low-cost, automated communication

Key Business Insights

- A 26.5% churn rate represents a significant, addressable revenue risk.
- Churn correlates with lower engagement, lower product diversity, and recent drops in spending — behavioral signals, not just demographics.
- A small “High Value” cluster drives a disproportionate share of revenue; retaining them is more valuable than acquiring new customers.
- Combining churn risk with CLV avoids two common mistakes: overspending on low-value customers and under-investing in at-risk high-value customers.
- Month-to-month contract customers are the highest-risk group and the best candidates for proactive retention offers (e.g. contract upgrade incentives).

Tech Stack

Data/ML: pandas, numpy, scikit-learn, XGBoost, LightGBM, lifetimes | Visualization: matplotlib, seaborn | Environment: Jupyter Notebook

Possible Extensions

- Hyperparameter tuning (grid/Bayesian search) for churn and CLV models.
- Survival-analysis-based CLV (BG/NBD + Gamma-Gamma) on real transactional history.
- Deploy the scoring pipeline as an API for real-time campaign targeting.
- A/B test the recommendation engine's retention offers against a control group.